VIDHYADEEP UNIVERSITY
Holy Flame Of Knowledge

Vidhyadeep Institute of Commerce & Information Technology, Anita (Kim)

PRACTICAL RECORD

ASP.NET

TY B.Sc. IT | Semester V

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INDEX

NO.	PROGRAM / PRACTICAL	PAGE	SIGN
BASIC ASP.NET PRACTICALS			
01	Create a simple ASP.NET page	2	
02	Code-behind coding	2	
03	Inline coding	3	
04	MSIL compilation	3	
05	CLR and exception handling	4	
06	Encapsulation	4	
07	Inheritance	5	
08	Method overloading	6	
09	Button click event	7	
10	Student information form	7	
ASSIGNMENT 2 - ASP.NET WEB FORMS			
01	SQL Server GridView data binding	10	
02	Login-form validators	10	
03	Master Page and student information	11	
04	Employee management	12	
05	Customer management	14	
06	Student management with Master Page	17	
07	Pharmacy medicine records	21	
08	Redirect from Login.aspx to Welcome.aspx	23	
09	Store and retrieve a cookie	24	
10	Login with Session and Cookie	25	

BASIC ASP.NET PRACTICALS

ASP.NET Foundation Practicals

01. Create a simple ASP.NET page

Aim: Create a simple ASP.NET PAGE.

Logic: A minimal Web Form renders a static server page.

PROGRAM / Default.aspx

```
<%@ Page Language="VB" %>
<!DOCTYPE html>
<html><head runat="server"><title>My First ASP.NET Page</title>
<link href="style.css" rel="stylesheet" />
</head><body><form id="form1" runat="server">
<main><h1>My First ASP.NET Page</h1>
<p>Hello from an ASP.NET Web Form.</p>
</main></form></body></html>
```

Common project files (reproduced in the appendix): style.css.

EXPECTED OUTPUT

```
My First ASP.NET Page - Hello from an ASP.NET Web Form.
```

Illustrative expected result based on source logic; ASP.NET execution was not available here.

Note: Classic ASP.NET Web Forms / VB.NET. Deploy this Program folder as a Web Site under .NET Framework 4.7.2; not intended for ASP.NET Core.

02. Code-behind coding

Aim: DEMONSTRATE CODE-BEHIND CODING.

Logic: Separate user interface (.aspx) from Visual Basic event logic (.aspx.vb).

PROGRAM / Default.aspx

```
<%@ Page Language="VB" AutoEventWireup="true" CodeFile="Default.aspx.vb" Inherits="CodeBehindDemo" %>
<!DOCTYPE html>
<html><head runat="server"><title>Code-behind Demo</title>
<link href="style.css" rel="stylesheet" />
</head><body><form id="form1" runat="server">
<main><h1>Code-behind Demo</h1>
<asp:Label ID="lblGreeting" runat="server" CssClass="message" />
</main></form></body></html>
```

PROGRAM / Default.aspx.vb

```
Partial Class CodeBehindDemo
    Inherits System.Web.UI.Page
    Protected Sub Page_Load(sender As Object, e As EventArgs)
        If Not IsPostBack Then
            lblGreeting.Text = "Hello from the VB code-behind file."
        End If
    End Sub
End Class
```

Common project files (reproduced in the appendix): style.css.

EXPECTED OUTPUT

```
Label: Hello from the VB code-behind file.
```

Illustrative expected result based on source logic; ASP.NET execution was not available here.

Note: Classic ASP.NET Web Forms / VB.NET. Deploy this Program folder as a Web Site under .NET Framework 4.7.2; not intended for ASP.NET Core.

03. Inline coding

Aim: Demonstrate inline coding.

Logic: Use an in-page server-side VB method to set a label.

PROGRAM / Default.aspx

```
<%@ Page Language="VB" %>
<script runat="server">
    Protected Sub Page_Load(sender As Object, e As EventArgs)
        If Not IsPostBack Then
            lblTime.Text = "Inline server-side code executed."
        End If
    End Sub
</script>
<!DOCTYPE html>
<html><head runat="server"><title>Inline VB</title>
<link href="style.css" rel="stylesheet" /></head>
<body><form id="form1" runat="server"><main>
<h1>Inline Code Demonstration</h1>
<asp:Label ID="lblTime" runat="server" CssClass="message" />
</main></form></body></html>
```

Common project files (reproduced in the appendix): style.css.

EXPECTED OUTPUT

```
Label: Inline server-side code executed.
```

Illustrative expected result based on source logic; ASP.NET execution was not available here.

Note: Classic ASP.NET Web Forms / VB.NET. Deploy this Program folder as a Web Site under .NET Framework 4.7.2; not intended for ASP.NET Core.

---

04. MSIL compilation

Aim: Demonstrate MSIL compilation.

Logic: Compile VB.NET to a managed assembly, then inspect the Intermediate Language using the Framework SDK.

PROGRAM / Build_And_Inspect.txt

```
Windows Developer Command Prompt with .NET Framework tools:
  vbc.exe /target:exe /out:Hello.exe Hello.vb
  ildasm.exe Hello.exe

Inspect the IL method body: ldstr, call, and ret are examples of IL
instructions. The CLR's JIT compiles IL to native instructions at run time.
The commands and IL disassembly are NOT runtime-verified in this environment.
```

PROGRAM / Hello.vb

```
Public Module Hello
    Public Sub Main()
        System.Console.WriteLine("Hello from managed .NET code")
    End Sub
End Module
```

Common project files (reproduced in the appendix): style.css.

EXPECTED OUTPUT

```
Expected console text: Hello from managed .NET code
```

Illustrative expected result based on source logic; ASP.NET execution was not available here.

Note: Classic ASP.NET Web Forms / VB.NET. Deploy this Program folder as a Web Site under .NET Framework 4.7.2; not intended for ASP.NET Core.

---

05. CLR and exception handling

Aim: Demonstrate CLR and Exception Handling.

Logic: ASP.NET code-behind catches a numerical format error without crashing the page.

PROGRAM / Default.aspx

```
<%@ Page Language="VB" AutoEventWireup="true" CodeFile="Default.aspx.vb" Inherits="ExceptionDemo" %>
<!DOCTYPE html>
<html><head runat="server"><title>Exception Handling</title>
<link href="style.css" rel="stylesheet" />
</head><body><form id="form1" runat="server">
<main><h1>Exception Handling</h1>
<label>Enter an integer</label>
<asp:TextBox ID="txtNumber" runat="server" />
<asp:Button ID="btnShow" runat="server" Text="Process" OnClick="btnShow_Click" />
<asp:Label ID="lblResult" runat="server" CssClass="message" />
</main></form></body></html>
```

PROGRAM / Default.aspx.vb

```
Partial Class ExceptionDemo
    Inherits System.Web.UI.Page
    Protected Sub btnShow_Click(sender As Object, e As EventArgs)
        Try
            Dim n As Integer = Integer.Parse(txtNumber.Text)
            lblResult.Text = "Square = " & (CLng(n) * n).ToString()
        Catch ex As FormatException
            lblResult.Text = "Please enter a valid integer."
        Catch ex As OverflowException
            lblResult.Text = "Number is outside the allowed range."
        End Try
    End Sub
End Class
```

Common project files (reproduced in the appendix): style.css.

EXPECTED OUTPUT

```
Input abc -> Please enter a valid integer.
```

Illustrative expected result based on source logic; ASP.NET execution was not available here.

Note: Classic ASP.NET Web Forms / VB.NET. Deploy this Program folder as a Web Site under .NET Framework 4.7.2; not intended for ASP.NET Core.

06. Encapsulation

Aim: Demonstrate Encapsulation.

Logic: A private field is exposed through a validated public property.

PROGRAM / App_Code/StudentProfile.vb

```
Public Class StudentProfile
    Private _name As String
    Public Property RollNo As Integer

    Public Property Name As String
        Get
            Return _name
        End Get
        Set(value As String)
            If String.IsNullOrWhiteSpace(value) Then
                Throw New ArgumentException("Name is required")
            End If
            _name = value.Trim()
        End Set
    End Property

    Public Function Describe() As String
        Return "Roll " & RollNo & " : " & Name
    End Function
End Class
```

PROGRAM / Default.aspx

```
<%@ Page Language="VB" AutoEventWireup="true" CodeFile="Default.aspx.vb" Inherits="EncapsulationDemo" %>
<!DOCTYPE html>
<html><head runat="server"><title>Encapsulation</title>
<link href="style.css" rel="stylesheet" />
</head><body><form id="form1" runat="server">
<main><h1>Encapsulation</h1>
<asp:Label ID="lblResult" runat="server" CssClass="message" />
</main></form></body></html>
```

PROGRAM / Default.aspx.vb

```
Partial Class EncapsulationDemo
    Inherits System.Web.UI.Page
    Protected Sub Page_Load(sender As Object, e As EventArgs)
        Dim student As New StudentProfile() With {.RollNo = 101}
        student.Name = "Aman"
        lblResult.Text = Server.HtmlEncode(student.Describe())
    End Sub
End Class
```

Common project files (reproduced in the appendix): style.css.

EXPECTED OUTPUT

```
Roll 101 : Aman
```

Illustrative expected result based on source logic; ASP.NET execution was not available here.

Note: Classic ASP.NET Web Forms / VB.NET. Deploy this Program folder as a Web Site under .NET Framework 4.7.2; not intended for ASP.NET Core.

07. Inheritance

Aim: Demonstrate Inheritance.

Logic: The derived class inherits state and overrides base-class behavior.

PROGRAM / App_Code/PersonBase.vb

```
Public Class PersonBase
    Public Property FullName As String
    Public Overridable Function Describe() As String
        Return "Person: " & FullName
    End Function
End Class
```

PROGRAM / App_Code/StudentRecord.vb

```
Public Class StudentRecord
    Inherits PersonBase
    Public Property Course As String
    Public Overrides Function Describe() As String
        Return "Student: " & FullName & " / " & Course
    End Function
End Class
```

PROGRAM / Default.aspx

```
<%@ Page Language="VB" AutoEventWireup="true" CodeFile="Default.aspx.vb" Inherits="InheritanceDemo" %>
<!DOCTYPE html>
<html><head runat="server"><title>Inheritance</title>
<link href="style.css" rel="stylesheet" />
</head><body><form id="form1" runat="server">
<main><h1>Inheritance</h1>
<asp:Label ID="lblResult" runat="server" CssClass="message" />
</main></form></body></html>
```

PROGRAM / Default.aspx.vb

```
Partial Class InheritanceDemo
    Inherits System.Web.UI.Page
    Protected Sub Page_Load(sender As Object, e As EventArgs)
        Dim student As New StudentRecord() With {
            .FullName = "Riya", .Course = "BSc.IT"
        }
        lblResult.Text = Server.HtmlEncode(student.Describe())
    End Sub
End Class
```

Common project files (reproduced in the appendix): style.css.

EXPECTED OUTPUT

```
Student: Riya / BSc.IT
```

Illustrative expected result based on source logic; ASP.NET execution was not available here.

Note: Classic ASP.NET Web Forms / VB.NET. Deploy this Program folder as a Web Site under .NET Framework 4.7.2; not intended for ASP.NET Core.

08. Method overloading

Aim: Demonstrate Method overloading.

Logic: Two Add methods share a name but accept different argument counts.

PROGRAM / App_Code/Calculator.vb

```
Public Class Calculator
    Public Function Add(a As Integer, b As Integer) As Integer
        Return a + b
    End Function
    Public Function Add(a As Integer, b As Integer,
                        c As Integer) As Integer
        Return a + b + c
    End Function
End Class
```

PROGRAM / Default.aspx

```
<%@ Page Language="VB" AutoEventWireup="true" CodeFile="Default.aspx.vb" Inherits="OverloadingDemo" %>
<!DOCTYPE html>
<html><head runat="server"><title>Method Overloading</title>
<link href="style.css" rel="stylesheet" />
</head><body><form id="form1" runat="server">
<main><h1>Method Overloading</h1>
<asp:Label ID="lblResult" runat="server" CssClass="message" />
</main></form></body></html>
```

PROGRAM / Default.aspx.vb

```
Partial Class OverloadingDemo
    Inherits System.Web.UI.Page
    Protected Sub Page_Load(sender As Object, e As EventArgs)
        Dim calc As New Calculator()
        lblResult.Text = "Two numbers: " & calc.Add(2, 3) &
            "; three numbers: " & calc.Add(2, 3, 4)
    End Sub
End Class
```

Common project files (reproduced in the appendix): style.css.

EXPECTED OUTPUT

```
Two numbers: 5; three numbers: 9
```

Illustrative expected result based on source logic; ASP.NET execution was not available here.

Note: Classic ASP.NET Web Forms / VB.NET. Deploy this Program folder as a Web Site under .NET Framework 4.7.2; not intended for ASP.NET Core.

09. Button click event

Aim: Demonstrate Button Click Event.

Logic: OnClick dispatches a server-side event handler on postback.

PROGRAM / Default.aspx

```
<%@ Page Language="VB" AutoEventWireup="true" CodeFile="Default.aspx.vb" Inherits="ButtonDemo" %>
<!DOCTYPE html>
<html><head runat="server"><title>Button Click Event</title>
<link href="style.css" rel="stylesheet" />
</head><body><form id="form1" runat="server">
<main><h1>Button Click Event</h1>
<asp:Button ID="btnWelcome" runat="server" Text="Click me"
    OnClick="btnWelcome_Click" />
<asp:Label ID="lblResult" runat="server" CssClass="message" />
</main></form></body></html>
```

PROGRAM / Default.aspx.vb

```
Partial Class ButtonDemo
    Inherits System.Web.UI.Page
    Protected Sub btnWelcome_Click(sender As Object, e As EventArgs)
        lblResult.Text = "Button clicked successfully."
    End Sub
End Class
```

Common project files (reproduced in the appendix): style.css.

EXPECTED OUTPUT

```
After click: Button clicked successfully.
```

Illustrative expected result based on source logic; ASP.NET execution was not available here.

Note: Classic ASP.NET Web Forms / VB.NET. Deploy this Program folder as a Web Site under .NET Framework 4.7.2; not intended for ASP.NET Core.

10. Student information form

Aim: Mini Practical: Student information form.

Logic: Collect and validate name, roll number, course and age; display safely encoded submission data.

PROGRAM / Default.aspx

```
<%@ Page Language="VB" AutoEventWireup="true" CodeFile="Default.aspx.vb" Inherits="StudentForm" %>
<!DOCTYPE html>
<html><head runat="server"><title>Student Information Form</title>
<link href="style.css" rel="stylesheet" />
</head><body><form id="form1" runat="server">
<main><h1>Student Information Form</h1>
<label>Name</label><asp:TextBox ID="txtName" runat="server" />
<asp:RequiredFieldValidator runat="server" ControlToValidate="txtName"
    ErrorMessage="Name required" CssClass="error" />
<label>Roll No</label><asp:TextBox ID="txtRoll" runat="server" />
<asp:RequiredFieldValidator runat="server" ControlToValidate="txtRoll"
    ErrorMessage="Roll no. required" CssClass="error" />
<label>Age</label><asp:TextBox ID="txtAge" runat="server" />
<asp:RangeValidator runat="server" ControlToValidate="txtAge"
    MinimumValue="1" MaximumValue="120" Type="Integer"
    ErrorMessage="Enter age 1-120" CssClass="error" />
<label>Course</label><asp:TextBox ID="txtCourse" runat="server" />
<asp:RequiredFieldValidator runat="server" ControlToValidate="txtCourse"
    ErrorMessage="Course required" CssClass="error" />
<br /><asp:Button ID="btnSave" runat="server" Text="Show student"
    OnClick="btnSave_Click" />
<asp:Label ID="lblResult" runat="server" CssClass="message" />
</main></form></body></html>
```

PROGRAM / Default.aspx.vb

```
Partial Class StudentForm
    Inherits System.Web.UI.Page
    Protected Sub btnSave_Click(sender As Object, e As EventArgs)
        If Not Page.IsValid Then Return
        Dim roll As Integer
        If Not Integer.TryParse(txtRoll.Text, roll) OrElse roll <= 0 Then
            lblResult.Text = "Roll number must be a positive integer."
            Return
        End If
        lblResult.Text = "Student: " & Server.HtmlEncode(txtName.Text) &
            " | Roll: " & roll & " | Age: " & Server.HtmlEncode(txtAge.Text) &
            " | Course: " & Server.HtmlEncode(txtCourse.Text)
    End Sub
End Class
```

Common project files (reproduced in the appendix): style.css.

EXPECTED OUTPUT

```
Example after submit: Student: Riya | Roll: 102 | Age: 20 | Course: BSc.IT
```

Illustrative expected result based on source logic; ASP.NET execution was not available here.

Note: Classic ASP.NET Web Forms / VB.NET. Deploy this Program folder as a Web Site under .NET Framework 4.7.2; not intended for ASP.NET Core.

ASSIGNMENT 2 - ASP.NET WEB FORMS

ASP.NET Web Forms

DATABASE SETUP / PracticalDB.sql

```
-- SQL Server / LocalDB. Run in SSMS or sqlcmd before testing database pages.
IF DB_ID('PracticalDB') IS NULL CREATE DATABASE PracticalDB;
GO
USE PracticalDB;
GO
IF OBJECT_ID('dbo.Students','U') IS NULL
CREATE TABLE dbo.Students (
  RollNo INT PRIMARY KEY, Name NVARCHAR(50) NOT NULL,
  Age INT NOT NULL CHECK (Age BETWEEN 1 AND 120),
  Gender NVARCHAR(10) NOT NULL, Course NVARCHAR(30) NOT NULL
);
IF OBJECT_ID('dbo.Employee','U') IS NULL
CREATE TABLE dbo.Employee (
  EmployeeID INT IDENTITY(1,1) PRIMARY KEY,
  EmployeeName NVARCHAR(50) NOT NULL, Department NVARCHAR(30) NOT NULL,
  Salary DECIMAL(10,2) NOT NULL CHECK (Salary >= 0),
  ContactNo NVARCHAR(15) NOT NULL
);
IF OBJECT_ID('dbo.Customer','U') IS NULL
CREATE TABLE dbo.Customer (
  CustomerID INT IDENTITY(1,1) PRIMARY KEY,
  CustomerName NVARCHAR(50) NOT NULL, Email NVARCHAR(60) NOT NULL,
  Phone NVARCHAR(15) NOT NULL, Address NVARCHAR(100) NOT NULL
);
IF OBJECT_ID('dbo.Medicine','U') IS NULL
CREATE TABLE dbo.Medicine (
  MedicineID INT IDENTITY(1,1) PRIMARY KEY,
  MedicineName NVARCHAR(50) NOT NULL, Category NVARCHAR(30) NOT NULL,
  Price DECIMAL(10,2) NOT NULL CHECK (Price >= 0),
  Quantity INT NOT NULL CHECK (Quantity >= 0), ExpiryDate DATE NOT NULL
);
GO
-- Demonstration rows only, inserted once when empty.
IF NOT EXISTS (SELECT 1 FROM dbo.Students)
 INSERT dbo.Students (RollNo,Name,Age,Gender,Course) VALUES
 (101,'Prince',19,'Male','BCA'),(102,'Karan',20,'Male','BCA'),
 (103,'Harshit',21,'Male','BBA'),(104,'Jeel',18,'Male','BA');
IF NOT EXISTS (SELECT 1 FROM dbo.Employee)
 INSERT dbo.Employee (EmployeeName,Department,Salary,ContactNo) VALUES
 ('Prince','IT',45000,'9568650011'),('Karan','HR',32000,'9876588022'),
 ('Harshit','Sales',38000,'9876567033');
IF NOT EXISTS (SELECT 1 FROM dbo.Customer)
 INSERT dbo.Customer (CustomerName,Email,Phone,Address) VALUES
 ('Kevin','kevin4@example.com','9876589044','Surat'),
 ('Ram','ram3@example.com','9876592033','Gandhinagar');
IF NOT EXISTS (SELECT 1 FROM dbo.Medicine)
 INSERT dbo.Medicine (MedicineName,Category,Price,Quantity,ExpiryDate) VALUES
 ('Cetirizine','Tablet',35.50,80,'2027-08-31'),
 ('Vitamin Syrup','Syrup',95.00,50,'2027-09-15');
GO
```

PROJECT SETUP / HOW_TO_RUN.md

```
# ASP.NET Assignment 2 - Windows setup

1. Use **Visual Studio 2022 with ASP.NET and web development** plus **.NET Framework 4.7.2** targeting
pack and SQL Server LocalDB.
2. Run `PracticalDB.sql` against `(LocalDB)\MSSQLLocalDB`.
3. Each `Program_XX` folder is an independent **ASP.NET Web Site**, not an ASP.NET Core app. Open the
desired folder with Visual Studio **File > Open > Web Site**.
4. Review `Web.config` and adapt the connection string to your actual SQL Server. The sample uses
Integrated Security.
5. Run with IIS Express; test CRUD, navigation, form validation, cookie/session and logout.
6. Programs 8 and 10 use hard-coded demo credentials `Prince` / `prince123`; these are teaching
demonstrations, **not secure production authentication**.
7. The deliverable's .NET compilation, SQL Server connection, and HTTP behavior are **not
runtime-verified** here because a Windows .NET Framework/SQL Server toolchain is unavailable.
```

01. SQL Server GridView data binding

Aim: Display student records stored in SQL Server dynamically through an ASP.NET GridView.

Logic: Fill a DataTable using ADO.NET on first page load and bind it to the GridView.

PROGRAM / Prog1.aspx

```
<%@ Page Language="VB" AutoEventWireup="true" CodeFile="Prog1.aspx.vb" Inherits="Prog1" %>
<!DOCTYPE html>
<html><head runat="server"><title>Student Records</title><link href="site.css" rel="stylesheet"
/></head><body><form id="form1" runat="server">
<div class="wrapper"><h2>Student Records</h2>
<asp:GridView ID="gvStudents" runat="server" CssClass="grid"
    AutoGenerateColumns="true" />
</div></form></body></html>
```

PROGRAM / Prog1.aspx.vb

```
Partial Class Prog1
    Inherits System.Web.UI.Page
    Protected Sub Page_Load(sender As Object, e As EventArgs)
        If Not IsPostBack Then
            gvStudents.DataSource = Db.ReadTable(
                "SELECT RollNo, Name, Age, Gender, Course FROM Students ORDER BY RollNo")
            gvStudents.DataBind()
        End If
    End Sub
End Class
```

Common project files (reproduced in the appendix): App_Code/Db.vb, site.css, Web.config.

EXPECTED OUTPUT

```
GridView headings: RollNo | Name | Age | Gender | Course
Example first row from supplied seed: 101 | Prince | 19 | Male | BCA
```

Illustrative expected output based on source; Web Forms/LocalDB runtime testing was not possible on this host.

Note: ASP.NET Web Forms (.NET Framework 4.7.2) with Visual Basic code-behind.

02. Login-form validators

Aim: Build Username, Password and Confirm Password fields with RequiredFieldValidator, CompareValidator and clear errors.

Logic: ASP.NET validates mandatory fields and password confirmation before the button handler runs.

PROGRAM / Prog2.aspx

```
<%@ Page Language="VB" AutoEventWireup="true" CodeFile="Prog2.aspx.vb" Inherits="Prog2" %>
<!DOCTYPE html>
<html><head runat="server"><title>Login Form Validation</title><link href="site.css" rel="stylesheet"
/></head><body><form id="form1" runat="server">
<div class="wrapper"><h2>Login Form Validation</h2>
<div class="field"><label>Username</label>
  <asp:TextBox ID="txtUser" runat="server" /></div>
<asp:RequiredFieldValidator runat="server" ControlToValidate="txtUser"
    ErrorMessage="Username is required" CssClass="error" Display="Dynamic" />
<div class="field"><label>Password</label>
  <asp:TextBox ID="txtPass" runat="server" TextMode="Password" /></div>
<asp:RequiredFieldValidator runat="server" ControlToValidate="txtPass"
    ErrorMessage="Password is required" CssClass="error" Display="Dynamic" />
<div class="field"><label>Confirm password</label>
  <asp:TextBox ID="txtConfirm" runat="server" TextMode="Password" /></div>
<asp:RequiredFieldValidator runat="server" ControlToValidate="txtConfirm"
    ErrorMessage="Confirmation is required" CssClass="error" Display="Dynamic" />
<asp:CompareValidator runat="server" ControlToValidate="txtConfirm"
    ControlToCompare="txtPass" Operator="Equal" Type="String"
    ErrorMessage="Passwords do not match" CssClass="error" Display="Dynamic" />
<br /><asp:Button ID="btnValidate" runat="server" Text="Validate"
    CssClass="action" OnClick="btnValidate_Click" />
<asp:Label ID="lblMessage" runat="server" CssClass="message" />
</div></form></body></html>
```

PROGRAM / Prog2.aspx.vb

```
Partial Class Prog2
    Inherits System.Web.UI.Page
    Protected Sub btnValidate_Click(sender As Object, e As EventArgs)
        If Page.IsValid Then
            lblMessage.Text = "Validation successful for " &
                Server.HtmlEncode(txtUser.Text) & "."
        End If
    End Sub
End Class
```

Common project files (reproduced in the appendix): site.css, Web.config.

EXPECTED OUTPUT

```
Empty Username -> Username is required
Different passwords -> Passwords do not match
Valid matching inputs -> Validation successful for Prince.
```

Illustrative expected output based on source; Web Forms/LocalDB runtime testing was not possible on this host.

Note: This exercise demonstrates input validation only; it does not authenticate a user against an identity provider.

03. Master Page and student information

Aim: Create a Master Page with Header, Navigation, Footer and ContentPlaceHolder; display student information in a Content Page.

Logic: Place the shared page structure in a .master file; define the content inside an asp:Content control.

PROGRAM / Prog3.aspx

```
<%@ Page Language="VB" AutoEventWireup="true" MasterPageFile="~/Student.master" CodeFile="Prog3.aspx.vb"
Inherits="Prog3" %>
<asp:Content ID="BodyContent" ContentPlaceHolderID="MainContent" runat="server">
<h2>Student Information</h2>
<p><strong>Name:</strong> <asp:Label ID="lblName" runat="server" /></p>
<p><strong>Roll no:</strong> <asp:Label ID="lblRoll" runat="server" /></p>
<p><strong>Course:</strong> <asp:Label ID="lblCourse" runat="server" /></p>
<p><strong>Semester:</strong> <asp:Label ID="lblSemester" runat="server" /></p>
</asp:Content>
```

PROGRAM / Prog3.aspx.vb

```
Partial Class Prog3
    Inherits System.Web.UI.Page
    Protected Sub Page_Load(sender As Object, e As EventArgs)
        If Not IsPostBack Then
            lblName.Text = "Prince"
            lblRoll.Text = "101"
            lblCourse.Text = "TY BCA"
            lblSemester.Text = "5"
        End If
    End Sub
End Class
```

PROGRAM / Student.master

```
<%@ Master Language="VB" %>
<!DOCTYPE html>
<html><head runat="server"><title>Student Portal</title>
<link href="site.css" rel="stylesheet" /></head><body>
<form id="form1" runat="server">
<header class="portal-head"><h1>Student Portal</h1></header>
<nav><a href="Prog3.aspx">Home</a>
     <a href="Prog3.aspx">Student Info</a>
     <a href="Prog3.aspx">About</a></nav>
<div class="wrapper">
<asp:ContentPlaceHolder ID="MainContent" runat="server" />
</div><footer>Vidhyadeep University · 2026</footer>
</form></body></html>
```

Common project files (reproduced in the appendix): site.css, Web.config.

EXPECTED OUTPUT

```
Student Portal / Home - Student Info - About
Name: Prince | Roll no: 101 | Course: TY BCA | Semester: 5
```

Illustrative expected output based on source; Web Forms/LocalDB runtime testing was not possible on this host.

Note: ASP.NET Web Forms (.NET Framework 4.7.2) with Visual Basic code-behind.

04. Employee management

Aim: Implement insert, view, update and delete for Employee using the prescribed fields: EmployeeName, Department, Salary, ContactNo and EmployeeID.

Logic: Display table rows in GridView; choose Select to load a record; use Insert, Update, Delete and Clear; rebind after changes.

PROGRAM / Prog4.aspx

```
<%@ Page Language="VB" AutoEventWireup="true" CodeFile="Prog4.aspx.vb" Inherits="Prog4" %>
<!DOCTYPE html>
<html><head runat="server"><title>Employee Records</title><link href="site.css" rel="stylesheet"
/></head><body><form id="form1" runat="server">
<div class="wrapper"><h2>Employee Records</h2>
<div class="field"><label>Employee name</label><asp:TextBox ID="txtEmployeeName" runat="server" /></div>
<div class="field"><label>Department</label><asp:TextBox ID="txtDepartment" runat="server" /></div>
<div class="field"><label>Salary</label><asp:TextBox ID="txtSalary" runat="server" /></div>
<div class="field"><label>Contact no.</label><asp:TextBox ID="txtContactNo" runat="server" /></div>
<asp:HiddenField ID="hfKey" runat="server" />
<asp:Button ID="btnAdd" runat="server" CssClass="action"
    Text="Insert" OnClick="btnAdd_Click" />
<asp:Button ID="btnUpdate" runat="server" CssClass="action"
    Text="Update" OnClick="btnUpdate_Click" />
<asp:Button ID="btnDelete" runat="server" CssClass="action"
    Text="Delete" OnClick="btnDelete_Click" />
<asp:Button ID="btnClear" runat="server" CssClass="action"
    Text="Clear" OnClick="btnClear_Click" />
<asp:Label ID="lblMessage" runat="server" CssClass="message" />
<asp:GridView ID="gvData" runat="server" CssClass="grid"
    AutoGenerateColumns="False" AutoGenerateSelectButton="True"
    DataKeyNames="EmployeeID" OnSelectedIndexChanged="gvData_SelectedIndexChanged">
  <Columns>
    <asp:BoundField DataField="EmployeeID" HeaderText="ID" />
    <asp:BoundField DataField="EmployeeName" HeaderText="Employee name" />
    <asp:BoundField DataField="Department" HeaderText="Department" />
    <asp:BoundField DataField="Salary" HeaderText="Salary" />
    <asp:BoundField DataField="ContactNo" HeaderText="Contact no." />
  </Columns>
</asp:GridView>

</div></form></body></html>
```

PROGRAM / Prog4.aspx.vb

```
Imports System
Imports System.Data
Imports System.Data.SqlClient
Imports System.Globalization

Partial Class Prog4
    Inherits System.Web.UI.Page

    Protected Sub Page_Load(sender As Object, e As EventArgs)
        If Not IsPostBack Then BindGrid()
    End Sub

    Private Sub BindGrid()
        gvData.DataSource = Db.ReadTable(
            "SELECT * FROM Employee ORDER BY EmployeeID")
        gvData.DataBind()
    End Sub

    Private Function FormParams() As SqlParameter()
        If String.IsNullOrWhiteSpace(txtEmployeeName.Text) Then
            Throw New ArgumentException("Employee name is required.")
        End If
        If String.IsNullOrWhiteSpace(txtDepartment.Text) Then
            Throw New ArgumentException("Department is required.")
        End If
        If String.IsNullOrWhiteSpace(txtSalary.Text) Then
            Throw New ArgumentException("Salary is required.")
        End If
        If String.IsNullOrWhiteSpace(txtContactNo.Text) Then
            Throw New ArgumentException("Contact no. is required.")
        End If
        Return New SqlParameter() {
            New SqlParameter("@EmployeeName", txtEmployeeName.Text.Trim()),
            New SqlParameter("@Department", txtDepartment.Text.Trim()),
            New SqlParameter("@Salary", Decimal.Parse(txtSalary.Text, CultureInfo.InvariantCulture)),
            New SqlParameter("@ContactNo", txtContactNo.Text.Trim())
        }
    End Function

    Private Sub ResetForm()
        txtEmployeeName.Text = ""
        txtDepartment.Text = ""
        txtSalary.Text = ""
        txtContactNo.Text = ""
        hfKey.Value = ""
        gvData.SelectedIndex = -1
    End Sub

    Protected Sub btnAdd_Click(sender As Object, e As EventArgs)
        Try
            Db.Execute("INSERT INTO Employee (EmployeeName,Department,Salary,ContactNo) " &
                "VALUES (@EmployeeName,@Department,@Salary,@ContactNo)", FormParams())
            lblMessage.Text = "Record inserted."
            ResetForm()
            BindGrid()
        Catch ex As Exception
            lblMessage.Text = "Insert failed: check all field values."
        End Try
    End Sub

    Protected Sub btnUpdate_Click(sender As Object, e As EventArgs)
        If hfKey.Value = "" Then
            lblMessage.Text = "Select a record first."
            Return
        End If
        Try
            Dim ps As New List(Of SqlParameter)(FormParams())
            ps.Add(New SqlParameter("@Key", Integer.Parse(hfKey.Value)))
            Db.Execute("UPDATE Employee SET EmployeeName=@EmployeeName,Department=@Department,Salary=@Sal
ary,ContactNo=@ContactNo WHERE EmployeeID=@Key", ps.ToArray())
            lblMessage.Text = "Record updated."
            ResetForm()
            BindGrid()
        Catch ex As Exception
```

```
            lblMessage.Text = "Update failed: check all field values."
        End Try
    End Sub

    Protected Sub btnDelete_Click(sender As Object, e As EventArgs)
        Dim key As Integer
        If Not Integer.TryParse(hfKey.Value, key) Then
            lblMessage.Text = "Select a record first."
            Return
        End If
        Db.Execute("DELETE FROM Employee WHERE EmployeeID=@Key",
            New SqlParameter("@Key", key))
        lblMessage.Text = "Record deleted."
        ResetForm()
        BindGrid()
    End Sub

    Protected Sub btnClear_Click(sender As Object, e As EventArgs)
        ResetForm()
        lblMessage.Text = ""
    End Sub

    Protected Sub gvData_SelectedIndexChanged(sender As Object, e As EventArgs)
        Dim key As Integer = Convert.ToInt32(gvData.SelectedDataKey.Value)
        Dim data As DataTable = Db.ReadTable(
            "SELECT * FROM Employee WHERE EmployeeID=@Key",
            New SqlParameter("@Key", key))
        If data.Rows.Count = 0 Then Return
        Dim row As DataRow = data.Rows(0)
        hfKey.Value = key.ToString()
        txtEmployeeName.Text = row("EmployeeName").ToString()
        txtDepartment.Text = row("Department").ToString()
        txtSalary.Text = row("Salary").ToString()
        txtContactNo.Text = row("ContactNo").ToString()
    End Sub
End Class
```

Common project files (reproduced in the appendix): App_Code/Db.vb, site.css, Web.config.

EXPECTED OUTPUT

```
Employee grid lists the SQL Server demo rows.
Insert -> Record inserted.  Select then Update -> Record updated.
Select then Delete -> Record deleted.
```

Illustrative expected output based on source; Web Forms/LocalDB runtime testing was not possible on this host.

Note: A SQL Server PracticalDB setup script and connection string are included. All queries use SQL parameters; server-side form conversion is validated by exception handling.

05. Customer management

Aim: Implement insert, view, update and delete for Customer using the prescribed fields: CustomerName, Email, Phone, Address and CustomerID.

Logic: Display table rows in GridView; choose Select to load a record; use Insert, Update, Delete and Clear; rebind after changes.

PROGRAM / Prog5.aspx

```
<%@ Page Language="VB" AutoEventWireup="true" CodeFile="Prog5.aspx.vb" Inherits="Prog5" %>
<!DOCTYPE html>
<html><head runat="server"><title>Customer Records</title><link href="site.css" rel="stylesheet"
/></head><body><form id="form1" runat="server">
<div class="wrapper"><h2>Customer Records</h2>
<div class="field"><label>Customer name</label><asp:TextBox ID="txtCustomerName" runat="server" /></div>
<div class="field"><label>Email</label><asp:TextBox ID="txtEmail" runat="server" /></div>
<div class="field"><label>Phone</label><asp:TextBox ID="txtPhone" runat="server" /></div>
<div class="field"><label>Address</label><asp:TextBox ID="txtAddress" runat="server" /></div>
<asp:HiddenField ID="hfKey" runat="server" />
<asp:Button ID="btnAdd" runat="server" CssClass="action"
    Text="Insert" OnClick="btnAdd_Click" />
<asp:Button ID="btnUpdate" runat="server" CssClass="action"
    Text="Update" OnClick="btnUpdate_Click" />
<asp:Button ID="btnDelete" runat="server" CssClass="action"
    Text="Delete" OnClick="btnDelete_Click" />
<asp:Button ID="btnClear" runat="server" CssClass="action"
    Text="Clear" OnClick="btnClear_Click" />
<asp:Label ID="lblMessage" runat="server" CssClass="message" />
<asp:GridView ID="gvData" runat="server" CssClass="grid"
    AutoGenerateColumns="False" AutoGenerateSelectButton="True"
    DataKeyNames="CustomerID" OnSelectedIndexChanged="gvData_SelectedIndexChanged">
  <Columns>
    <asp:BoundField DataField="CustomerID" HeaderText="ID" />
    <asp:BoundField DataField="CustomerName" HeaderText="Customer name" />
    <asp:BoundField DataField="Email" HeaderText="Email" />
    <asp:BoundField DataField="Phone" HeaderText="Phone" />
    <asp:BoundField DataField="Address" HeaderText="Address" />
  </Columns>
</asp:GridView>

</div></form></body></html>
```

PROGRAM / Prog5.aspx.vb

```
Imports System
Imports System.Data
Imports System.Data.SqlClient
Imports System.Globalization

Partial Class Prog5
    Inherits System.Web.UI.Page

    Protected Sub Page_Load(sender As Object, e As EventArgs)
        If Not IsPostBack Then BindGrid()
    End Sub

    Private Sub BindGrid()
        gvData.DataSource = Db.ReadTable(
            "SELECT * FROM Customer ORDER BY CustomerID")
        gvData.DataBind()
    End Sub

    Private Function FormParams() As SqlParameter()
        If String.IsNullOrWhiteSpace(txtCustomerName.Text) Then
            Throw New ArgumentException("Customer name is required.")
        End If
        If String.IsNullOrWhiteSpace(txtEmail.Text) Then
            Throw New ArgumentException("Email is required.")
        End If
        If String.IsNullOrWhiteSpace(txtPhone.Text) Then
            Throw New ArgumentException("Phone is required.")
        End If
        If String.IsNullOrWhiteSpace(txtAddress.Text) Then
            Throw New ArgumentException("Address is required.")
        End If
        Return New SqlParameter() {
            New SqlParameter("@CustomerName", txtCustomerName.Text.Trim()),
            New SqlParameter("@Email", txtEmail.Text.Trim()),
            New SqlParameter("@Phone", txtPhone.Text.Trim()),
            New SqlParameter("@Address", txtAddress.Text.Trim())
        }
    End Function

    Private Sub ResetForm()
        txtCustomerName.Text = ""
        txtEmail.Text = ""
        txtPhone.Text = ""
        txtAddress.Text = ""
        hfKey.Value = ""
        gvData.SelectedIndex = -1
    End Sub

    Protected Sub btnAdd_Click(sender As Object, e As EventArgs)
        Try
            Db.Execute("INSERT INTO Customer (CustomerName,Email,Phone,Address) " &
                "VALUES (@CustomerName,@Email,@Phone,@Address)", FormParams())
            lblMessage.Text = "Record inserted."
            ResetForm()
            BindGrid()
        Catch ex As Exception
            lblMessage.Text = "Insert failed: check all field values."
        End Try
    End Sub

    Protected Sub btnUpdate_Click(sender As Object, e As EventArgs)
        If hfKey.Value = "" Then
            lblMessage.Text = "Select a record first."
            Return
        End If
        Try
            Dim ps As New List(Of SqlParameter)(FormParams())
            ps.Add(New SqlParameter("@Key", Integer.Parse(hfKey.Value)))
            Db.Execute("UPDATE Customer SET CustomerName=@CustomerName,Email=@Email,Phone=@Phone,Address=
@Address WHERE CustomerID=@Key", ps.ToArray())
            lblMessage.Text = "Record updated."
            ResetForm()
            BindGrid()
        Catch ex As Exception
```

```
            lblMessage.Text = "Update failed: check all field values."
        End Try
    End Sub

    Protected Sub btnDelete_Click(sender As Object, e As EventArgs)
        Dim key As Integer
        If Not Integer.TryParse(hfKey.Value, key) Then
            lblMessage.Text = "Select a record first."
            Return
        End If
        Db.Execute("DELETE FROM Customer WHERE CustomerID=@Key",
            New SqlParameter("@Key", key))
        lblMessage.Text = "Record deleted."
        ResetForm()
        BindGrid()
    End Sub

    Protected Sub btnClear_Click(sender As Object, e As EventArgs)
        ResetForm()
        lblMessage.Text = ""
    End Sub

    Protected Sub gvData_SelectedIndexChanged(sender As Object, e As EventArgs)
        Dim key As Integer = Convert.ToInt32(gvData.SelectedDataKey.Value)
        Dim data As DataTable = Db.ReadTable(
            "SELECT * FROM Customer WHERE CustomerID=@Key",
            New SqlParameter("@Key", key))
        If data.Rows.Count = 0 Then Return
        Dim row As DataRow = data.Rows(0)
        hfKey.Value = key.ToString()
        txtCustomerName.Text = row("CustomerName").ToString()
        txtEmail.Text = row("Email").ToString()
        txtPhone.Text = row("Phone").ToString()
        txtAddress.Text = row("Address").ToString()
    End Sub
End Class
```

Common project files (reproduced in the appendix): App_Code/Db.vb, site.css, Web.config.

EXPECTED OUTPUT

```
Customer grid lists the SQL Server demo rows.
Insert -> Record inserted.  Select then Update -> Record updated.
Select then Delete -> Record deleted.
```

Illustrative expected output based on source; Web Forms/LocalDB runtime testing was not possible on this host.

Note: A SQL Server PracticalDB setup script and connection string are included. All queries use SQL parameters; server-side form conversion is validated by exception handling.

06. Student management with Master Page

Aim: Implement insert, view, update and delete for Students using the prescribed fields: Name, Age, Gender, Course and RollNo.

Logic: Display table rows in GridView; choose Select to load a record; use Insert, Update, Delete and Clear; rebind after changes.

PROGRAM / Prog6.aspx

```
<%@ Page Language="VB" AutoEventWireup="true" MasterPageFile="~/Student.master" CodeFile="Prog6.aspx.vb"
Inherits="Prog6" %>
<asp:Content ID="BodyContent" ContentPlaceHolderID="MainContent" runat="server">
<h2>Students Records</h2>
<div class="field"><label>Roll no.</label><asp:TextBox ID="txtRollNo" runat="server" /></div>
<div class="field"><label>Student name</label><asp:TextBox ID="txtName" runat="server" /></div>
<div class="field"><label>Age</label><asp:TextBox ID="txtAge" runat="server" /></div>
<div class="field"><label>Gender</label><asp:DropDownList ID="txtGender" runat="server"><asp:ListItem>Mal
e</asp:ListItem><asp:ListItem>Female</asp:ListItem><asp:ListItem>Other</asp:ListItem></asp:DropDownList><
/div>
<div class="field"><label>Course</label><asp:TextBox ID="txtCourse" runat="server" /></div>
<asp:HiddenField ID="hfKey" runat="server" />
<asp:Button ID="btnAdd" runat="server" CssClass="action"
    Text="Insert" OnClick="btnAdd_Click" />
<asp:Button ID="btnUpdate" runat="server" CssClass="action"
    Text="Update" OnClick="btnUpdate_Click" />
<asp:Button ID="btnDelete" runat="server" CssClass="action"
    Text="Delete" OnClick="btnDelete_Click" />
<asp:Button ID="btnClear" runat="server" CssClass="action"
    Text="Clear" OnClick="btnClear_Click" />
<asp:Label ID="lblMessage" runat="server" CssClass="message" />
<asp:GridView ID="gvData" runat="server" CssClass="grid"
    AutoGenerateColumns="False" AutoGenerateSelectButton="True"
    DataKeyNames="RollNo" OnSelectedIndexChanged="gvData_SelectedIndexChanged">
  <Columns>
    <asp:BoundField DataField="RollNo" HeaderText="ID" />
    <asp:BoundField DataField="Name" HeaderText="Student name" />
    <asp:BoundField DataField="Age" HeaderText="Age" />
    <asp:BoundField DataField="Gender" HeaderText="Gender" />
    <asp:BoundField DataField="Course" HeaderText="Course" />
  </Columns>
</asp:GridView>

</asp:Content>
```

PROGRAM / Prog6.aspx.vb

```
Imports System
Imports System.Data
Imports System.Data.SqlClient
Imports System.Globalization

Partial Class Prog6
    Inherits System.Web.UI.Page

    Protected Sub Page_Load(sender As Object, e As EventArgs)
        If Not IsPostBack Then BindGrid()
    End Sub

    Private Sub BindGrid()
        gvData.DataSource = Db.ReadTable(
            "SELECT * FROM Students ORDER BY RollNo")
        gvData.DataBind()
    End Sub

    Private Function FormParams() As SqlParameter()
        If String.IsNullOrWhiteSpace(txtRollNo.Text) Then
            Throw New ArgumentException("Roll no. is required.")
        End If
        If String.IsNullOrWhiteSpace(txtName.Text) Then
            Throw New ArgumentException("Student name is required.")
        End If
        If String.IsNullOrWhiteSpace(txtAge.Text) Then
            Throw New ArgumentException("Age is required.")
        End If
        If String.IsNullOrWhiteSpace(txtGender.SelectedValue) Then
            Throw New ArgumentException("Gender is required.")
        End If
        If String.IsNullOrWhiteSpace(txtCourse.Text) Then
            Throw New ArgumentException("Course is required.")
        End If
        Return New SqlParameter() {
            New SqlParameter("@RollNo", Integer.Parse(txtRollNo.Text, CultureInfo.InvariantCulture)),
            New SqlParameter("@Name", txtName.Text.Trim()),
            New SqlParameter("@Age", Integer.Parse(txtAge.Text, CultureInfo.InvariantCulture)),
            New SqlParameter("@Gender", txtGender.SelectedValue),
            New SqlParameter("@Course", txtCourse.Text.Trim())
        }
    End Function

    Private Sub ResetForm()
        txtRollNo.Text = ""
        txtName.Text = ""
        txtAge.Text = ""
        txtGender.SelectedIndex = 0
        txtCourse.Text = ""
        txtRollNo.ReadOnly = False
        hfKey.Value = ""
        gvData.SelectedIndex = -1
    End Sub

    Protected Sub btnAdd_Click(sender As Object, e As EventArgs)
        Try
            Db.Execute("INSERT INTO Students (RollNo,Name,Age,Gender,Course) " &
                "VALUES (@RollNo,@Name,@Age,@Gender,@Course)", FormParams())
            lblMessage.Text = "Record inserted."
            ResetForm()
            BindGrid()
        Catch ex As Exception
            lblMessage.Text = "Insert failed: check all field values."
        End Try
    End Sub

    Protected Sub btnUpdate_Click(sender As Object, e As EventArgs)
        If hfKey.Value = "" Then
            lblMessage.Text = "Select a record first."
            Return
        End If
        Try
            Dim ps As New List(Of SqlParameter)(FormParams())
            ps.Add(New SqlParameter("@Key", Integer.Parse(hfKey.Value)))
```

```
            Db.Execute("UPDATE Students SET Name=@Name,Age=@Age,Gender=@Gender,Course=@Course WHERE
RollNo=@Key", ps.ToArray())
            lblMessage.Text = "Record updated."
            ResetForm()
            BindGrid()
        Catch ex As Exception
            lblMessage.Text = "Update failed: check all field values."
        End Try
    End Sub

    Protected Sub btnDelete_Click(sender As Object, e As EventArgs)
        Dim key As Integer
        If Not Integer.TryParse(hfKey.Value, key) Then
            lblMessage.Text = "Select a record first."
            Return
        End If
        Db.Execute("DELETE FROM Students WHERE RollNo=@Key",
            New SqlParameter("@Key", key))
        lblMessage.Text = "Record deleted."
        ResetForm()
        BindGrid()
    End Sub

    Protected Sub btnClear_Click(sender As Object, e As EventArgs)
        ResetForm()
        lblMessage.Text = ""
    End Sub

    Protected Sub gvData_SelectedIndexChanged(sender As Object, e As EventArgs)
        Dim key As Integer = Convert.ToInt32(gvData.SelectedDataKey.Value)
        Dim data As DataTable = Db.ReadTable(
            "SELECT * FROM Students WHERE RollNo=@Key",
            New SqlParameter("@Key", key))
        If data.Rows.Count = 0 Then Return
        Dim row As DataRow = data.Rows(0)
        hfKey.Value = key.ToString()
        txtRollNo.Text = row("RollNo").ToString()
        txtName.Text = row("Name").ToString()
        txtAge.Text = row("Age").ToString()
        txtGender.SelectedValue = row("Gender").ToString()
        txtCourse.Text = row("Course").ToString()
        txtRollNo.ReadOnly = True
    End Sub
End Class
```

PROGRAM / Student.master

```
<%@ Master Language="VB" %>
<!DOCTYPE html>
<html><head runat="server"><title>College Portal</title>
<link href="site.css" rel="stylesheet" /></head><body>
<form id="form1" runat="server">
<header class="portal-head"><h1>College Portal</h1></header>
<nav><a href="Prog6.aspx">Home</a>
     <a href="Prog6.aspx">Students</a>
     <a href="Prog6.aspx">Contact</a></nav>
<div class="wrapper">
<asp:ContentPlaceHolder ID="MainContent" runat="server" />
</div><footer>Vidhyadeep University · 2026</footer>
</form></body></html>
```

Common project files (reproduced in the appendix): App_Code/Db.vb, site.css, Web.config.

EXPECTED OUTPUT

```
Students grid lists the SQL Server demo rows.
Insert -> Record inserted.  Select then Update -> Record updated.
Select then Delete -> Record deleted.
```

Illustrative expected output based on source; Web Forms/LocalDB runtime testing was not possible on this host.

Note: A SQL Server PracticalDB setup script and connection string are included. All queries use SQL parameters; server-side form conversion is validated by exception handling. The page uses a Master Page containing header, horizontal navigation, footer and ContentPlaceHolder.

07. Pharmacy medicine records

Aim: Implement insert, view, update and delete for Medicine using the prescribed fields: MedicineName, Category, Price, Quantity, ExpiryDate and MedicineID.

Logic: Display table rows in GridView; choose Select to load a record; use Insert, Update, Delete and Clear; rebind after changes.

PROGRAM / Prog7.aspx

```
<%@ Page Language="VB" AutoEventWireup="true" CodeFile="Prog7.aspx.vb" Inherits="Prog7" %>
<!DOCTYPE html>
<html><head runat="server"><title>Medicine Records</title><link href="site.css" rel="stylesheet"
/></head><body><form id="form1" runat="server">
<div class="wrapper"><h2>Medicine Records</h2>
<div class="field"><label>Medicine name</label><asp:TextBox ID="txtMedicineName" runat="server" /></div>
<div class="field"><label>Category</label><asp:TextBox ID="txtCategory" runat="server" /></div>
<div class="field"><label>Price</label><asp:TextBox ID="txtPrice" runat="server" /></div>
<div class="field"><label>Quantity</label><asp:TextBox ID="txtQuantity" runat="server" /></div>
<div class="field"><label>Expiry date</label><asp:TextBox ID="txtExpiryDate" runat="server"
TextMode="Date" /></div>
<asp:HiddenField ID="hfKey" runat="server" />
<asp:Button ID="btnAdd" runat="server" CssClass="action"
    Text="Insert" OnClick="btnAdd_Click" />
<asp:Button ID="btnUpdate" runat="server" CssClass="action"
    Text="Update" OnClick="btnUpdate_Click" />
<asp:Button ID="btnDelete" runat="server" CssClass="action"
    Text="Delete" OnClick="btnDelete_Click" />
<asp:Button ID="btnClear" runat="server" CssClass="action"
    Text="Clear" OnClick="btnClear_Click" />
<asp:Label ID="lblMessage" runat="server" CssClass="message" />
<asp:GridView ID="gvData" runat="server" CssClass="grid"
    AutoGenerateColumns="False" AutoGenerateSelectButton="True"
    DataKeyNames="MedicineID" OnSelectedIndexChanged="gvData_SelectedIndexChanged">
  <Columns>
    <asp:BoundField DataField="MedicineID" HeaderText="ID" />
    <asp:BoundField DataField="MedicineName" HeaderText="Medicine name" />
    <asp:BoundField DataField="Category" HeaderText="Category" />
    <asp:BoundField DataField="Price" HeaderText="Price" />
    <asp:BoundField DataField="Quantity" HeaderText="Quantity" />
    <asp:BoundField DataField="ExpiryDate" HeaderText="Expiry date" DataFormatString="{0:yyyy-MM-dd}" />
  </Columns>
</asp:GridView>

</div></form></body></html>
```

PROGRAM / Prog7.aspx.vb

```
Imports System
Imports System.Data
Imports System.Data.SqlClient
Imports System.Globalization

Partial Class Prog7
    Inherits System.Web.UI.Page

    Protected Sub Page_Load(sender As Object, e As EventArgs)
        If Not IsPostBack Then BindGrid()
    End Sub

    Private Sub BindGrid()
        gvData.DataSource = Db.ReadTable(
            "SELECT * FROM Medicine ORDER BY MedicineID")
        gvData.DataBind()
    End Sub

    Private Function FormParams() As SqlParameter()
        If String.IsNullOrWhiteSpace(txtMedicineName.Text) Then
            Throw New ArgumentException("Medicine name is required.")
        End If
        If String.IsNullOrWhiteSpace(txtCategory.Text) Then
            Throw New ArgumentException("Category is required.")
        End If
        If String.IsNullOrWhiteSpace(txtPrice.Text) Then
            Throw New ArgumentException("Price is required.")
        End If
        If String.IsNullOrWhiteSpace(txtQuantity.Text) Then
            Throw New ArgumentException("Quantity is required.")
        End If
        If String.IsNullOrWhiteSpace(txtExpiryDate.Text) Then
            Throw New ArgumentException("Expiry date is required.")
        End If
        Return New SqlParameter() {
            New SqlParameter("@MedicineName", txtMedicineName.Text.Trim()),
            New SqlParameter("@Category", txtCategory.Text.Trim()),
            New SqlParameter("@Price", Decimal.Parse(txtPrice.Text, CultureInfo.InvariantCulture)),
            New SqlParameter("@Quantity", Integer.Parse(txtQuantity.Text, CultureInfo.InvariantCulture)),
            New SqlParameter("@ExpiryDate", DateTime.ParseExact(txtExpiryDate.Text, "yyyy-MM-dd",
CultureInfo.InvariantCulture))
        }
    End Function

    Private Sub ResetForm()
        txtMedicineName.Text = ""
        txtCategory.Text = ""
        txtPrice.Text = ""
        txtQuantity.Text = ""
        txtExpiryDate.Text = ""
        hfKey.Value = ""
        gvData.SelectedIndex = -1
    End Sub

    Protected Sub btnAdd_Click(sender As Object, e As EventArgs)
        Try
            Db.Execute("INSERT INTO Medicine (MedicineName,Category,Price,Quantity,ExpiryDate) " &
                "VALUES (@MedicineName,@Category,@Price,@Quantity,@ExpiryDate)", FormParams())
            lblMessage.Text = "Record inserted."
            ResetForm()
            BindGrid()
        Catch ex As Exception
            lblMessage.Text = "Insert failed: check all field values."
        End Try
    End Sub

    Protected Sub btnUpdate_Click(sender As Object, e As EventArgs)
        If hfKey.Value = "" Then
            lblMessage.Text = "Select a record first."
            Return
        End If
        Try
            Dim ps As New List(Of SqlParameter)(FormParams())
            ps.Add(New SqlParameter("@Key", Integer.Parse(hfKey.Value)))
```

```
            Db.Execute("UPDATE Medicine SET MedicineName=@MedicineName,Category=@Category,Price=@Price,Qu
antity=@Quantity,ExpiryDate=@ExpiryDate WHERE MedicineID=@Key", ps.ToArray())
            lblMessage.Text = "Record updated."
            ResetForm()
            BindGrid()
        Catch ex As Exception
            lblMessage.Text = "Update failed: check all field values."
        End Try
    End Sub

    Protected Sub btnDelete_Click(sender As Object, e As EventArgs)
        Dim key As Integer
        If Not Integer.TryParse(hfKey.Value, key) Then
            lblMessage.Text = "Select a record first."
            Return
        End If
        Db.Execute("DELETE FROM Medicine WHERE MedicineID=@Key",
            New SqlParameter("@Key", key))
        lblMessage.Text = "Record deleted."
        ResetForm()
        BindGrid()
    End Sub

    Protected Sub btnClear_Click(sender As Object, e As EventArgs)
        ResetForm()
        lblMessage.Text = ""
    End Sub

    Protected Sub gvData_SelectedIndexChanged(sender As Object, e As EventArgs)
        Dim key As Integer = Convert.ToInt32(gvData.SelectedDataKey.Value)
        Dim data As DataTable = Db.ReadTable(
            "SELECT * FROM Medicine WHERE MedicineID=@Key",
            New SqlParameter("@Key", key))
        If data.Rows.Count = 0 Then Return
        Dim row As DataRow = data.Rows(0)
        hfKey.Value = key.ToString()
        txtMedicineName.Text = row("MedicineName").ToString()
        txtCategory.Text = row("Category").ToString()
        txtPrice.Text = row("Price").ToString()
        txtQuantity.Text = row("Quantity").ToString()
        txtExpiryDate.Text = CDate(row("ExpiryDate")).ToString("yyyy-MM-dd")
    End Sub
End Class
```

Common project files (reproduced in the appendix): App_Code/Db.vb, site.css, Web.config.

EXPECTED OUTPUT

```
Medicine grid lists the SQL Server demo rows.
Insert -> Record inserted.  Select then Update -> Record updated.
Select then Delete -> Record deleted.
```

Illustrative expected output based on source; Web Forms/LocalDB runtime testing was not possible on this host.

Note: A SQL Server PracticalDB setup script and connection string are included. All queries use SQL parameters; server-side form conversion is validated by exception handling.

08. Redirect from Login.aspx to Welcome.aspx

Aim: On successful login, redirect from Login.aspx to Welcome.aspx using Response.Redirect().

Logic: Validate classroom demo credentials and store the selected username in Session before redirecting.

PROGRAM / Login.aspx

```
<%@ Page Language="VB" AutoEventWireup="true" CodeFile="Login.aspx.vb" Inherits="Login" %>
<!DOCTYPE html>
<html><head runat="server"><title>Login and Redirect</title><link href="site.css" rel="stylesheet"
/></head><body><form id="form1" runat="server">
<div class="wrapper"><h2>Login and Redirect</h2>
<div class="field"><label>Username</label><asp:TextBox ID="txtUser" runat="server" /></div>
<div class="field"><label>Password</label><asp:TextBox ID="txtPass" runat="server" TextMode="Password"
/></div>
<asp:Button ID="btnLogin" runat="server" Text="Login" CssClass="action" OnClick="btnLogin_Click" />
<asp:Label ID="lblMessage" runat="server" CssClass="error" />
</div></form></body></html>
```

PROGRAM / Login.aspx.vb

```
Partial Class Login
    Inherits System.Web.UI.Page
    Protected Sub btnLogin_Click(sender As Object, e As EventArgs)
        If txtUser.Text = "Prince" AndAlso txtPass.Text = "prince123" Then
            Session("redirectUser") = txtUser.Text
            Response.Redirect("Welcome.aspx")
        Else
            lblMessage.Text = "Invalid username or password."
        End If
    End Sub
End Class
```

PROGRAM / Welcome.aspx

```
<%@ Page Language="VB" AutoEventWireup="true" CodeFile="Welcome.aspx.vb" Inherits="Welcome" %>
<!DOCTYPE html>
<html><head runat="server"><title>Welcome</title><link href="site.css" rel="stylesheet"
/></head><body><form id="form1" runat="server">
<div class="wrapper"><h2>Welcome</h2>
<asp:Label ID="lblWelcome" runat="server" CssClass="message" />
<a href="Login.aspx">Back to login</a>
</div></form></body></html>
```

PROGRAM / Welcome.aspx.vb

```
Partial Class Welcome
    Inherits System.Web.UI.Page
    Protected Sub Page_Load(sender As Object, e As EventArgs)
        If Session("redirectUser") Is Nothing Then
            Response.Redirect("Login.aspx")
            Return
        End If
        lblWelcome.Text = "Welcome, " &
            Server.HtmlEncode(Session("redirectUser").ToString()) & "!"
    End Sub
End Class
```

Common project files (reproduced in the appendix): site.css, Web.config.

EXPECTED OUTPUT

```
Login Prince / prince123 -> navigates to Welcome.aspx
Welcome, Prince!
Invalid credentials -> error message.
```

Illustrative expected output based on source; Web Forms/LocalDB runtime testing was not possible on this host.

Note: Credentials are hard-coded solely for the classroom demo; replace them with secure authentication in real deployments.

---

09. Store and retrieve a cookie

Aim: Accept a name, save it in a Cookie; on another page read the cookie and display the name.

Logic: Write a URL-encoded HttpOnly cookie with one-day expiry; HTML-encode the decoded value when displaying it.

PROGRAM / GetCookie.aspx

```
<%@ Page Language="VB" AutoEventWireup="true" CodeFile="GetCookie.aspx.vb" Inherits="GetCookie" %>
<!DOCTYPE html>
<html><head runat="server"><title>Read Name from Cookie</title><link href="site.css" rel="stylesheet"
/></head><body><form id="form1" runat="server">
<div class="wrapper"><h2>Read Name from Cookie</h2>
<asp:Label ID="lblResult" runat="server" CssClass="message" />
<p><a href="SetCookie.aspx">Back</a></p>
</div></form></body></html>
```

PROGRAM / GetCookie.aspx.vb

```
Partial Class GetCookie
    Inherits System.Web.UI.Page
    Protected Sub Page_Load(sender As Object, e As EventArgs)
        Dim item As HttpCookie = Request.Cookies("UserName")
        lblResult.Text = If(item Is Nothing,
            "No cookie found.",
            "Hello, " & Server.HtmlEncode(Server.UrlDecode(item.Value)))
    End Sub
End Class
```

PROGRAM / SetCookie.aspx

```
<%@ Page Language="VB" AutoEventWireup="true" CodeFile="SetCookie.aspx.vb" Inherits="SetCookie" %>
<!DOCTYPE html>
<html><head runat="server"><title>Store Name in Cookie</title><link href="site.css" rel="stylesheet"
/></head><body><form id="form1" runat="server">
<div class="wrapper"><h2>Store Name in Cookie</h2>
<div class="field"><label>Name</label><asp:TextBox ID="txtName" runat="server" /></div>
<asp:Button ID="btnSave" runat="server" Text="Save cookie" CssClass="action" OnClick="btnSave_Click" />
<asp:Label ID="lblMessage" runat="server" CssClass="message" />
<p><a href="GetCookie.aspx">Open cookie read page</a></p>
</div></form></body></html>
```

PROGRAM / SetCookie.aspx.vb

```
Partial Class SetCookie
    Inherits System.Web.UI.Page
    Protected Sub btnSave_Click(sender As Object, e As EventArgs)
        If String.IsNullOrWhiteSpace(txtName.Text) Then
            lblMessage.Text = "Please provide a name."
            Return
        End If
        Dim cookie As New HttpCookie("UserName", Server.UrlEncode(txtName.Text))
        cookie.Expires = DateTime.Now.AddDays(1)
        cookie.HttpOnly = True
        cookie.Secure = Request.IsSecureConnection
        Response.Cookies.Add(cookie)
        lblMessage.Text = "Name stored in cookie."
    End Sub
End Class
```

Common project files (reproduced in the appendix): site.css, Web.config.

EXPECTED OUTPUT

```
Save name Prince -> Name stored in cookie.
Read page -> Hello, Prince
```

Illustrative expected output based on source; Web Forms/LocalDB runtime testing was not possible on this host.

Note: ASP.NET Web Forms (.NET Framework 4.7.2) with Visual Basic code-behind.

---

10. Login with Session and Cookie

Aim: Create login with Username/Password; store username in Session and Cookie, redirect to Welcome page, display both, and provide Logout to clear Session.

Logic: Use Session to authorize access; cookie is only a demonstration of client-side state. Clear both on logout.

PROGRAM / LoginSession.aspx

```
<%@ Page Language="VB" AutoEventWireup="true" CodeFile="LoginSession.aspx.vb" Inherits="LoginSession" %>
<!DOCTYPE html>
<html><head runat="server"><title>Session Login</title><link href="site.css" rel="stylesheet"
/></head><body><form id="form1" runat="server">
<div class="wrapper"><h2>Session Login</h2>
<div class="field"><label>Username</label><asp:TextBox ID="txtUser" runat="server" /></div>
<div class="field"><label>Password</label><asp:TextBox ID="txtPass" runat="server" TextMode="Password"
/></div>
<asp:Button ID="btnLogin" runat="server" Text="Login" CssClass="action" OnClick="btnLogin_Click" />
<asp:Label ID="lblMessage" runat="server" CssClass="error" />
</div></form></body></html>
```

PROGRAM / LoginSession.aspx.vb

```
Partial Class LoginSession
    Inherits System.Web.UI.Page
    Protected Sub btnLogin_Click(sender As Object, e As EventArgs)
        If txtUser.Text <> "Prince" OrElse txtPass.Text <> "prince123" Then
            lblMessage.Text = "Invalid username or password."
            Return
        End If
        Session("user") = txtUser.Text
        Dim cookie As New HttpCookie("user", Server.UrlEncode(txtUser.Text))
        cookie.Expires = DateTime.Now.AddDays(1)
        cookie.HttpOnly = True
        cookie.Secure = Request.IsSecureConnection
        Response.Cookies.Add(cookie)
        Response.Redirect("WelcomeSession.aspx")
    End Sub
End Class
```

PROGRAM / WelcomeSession.aspx

```
<%@ Page Language="VB" AutoEventWireup="true" CodeFile="WelcomeSession.aspx.vb" Inherits="WelcomeSession"
%>
<!DOCTYPE html>
<html><head runat="server"><title>Welcome - Session and Cookie</title><link href="site.css"
rel="stylesheet" /></head><body><form id="form1" runat="server">
<div class="wrapper"><h2>Welcome - Session and Cookie</h2>
<p>From Session: <asp:Label ID="lblSession" runat="server" /></p>
<p>From Cookie: <asp:Label ID="lblCookie" runat="server" /></p>
<asp:Button ID="btnLogout" runat="server" Text="Logout" CssClass="action"
    OnClick="btnLogout_Click" />
</div></form></body></html>
```

PROGRAM / WelcomeSession.aspx.vb

```
Partial Class WelcomeSession
    Inherits System.Web.UI.Page
    Protected Sub Page_Load(sender As Object, e As EventArgs)
        If Session("user") Is Nothing Then
            Response.Redirect("LoginSession.aspx")
            Return
        End If
        lblSession.Text = Server.HtmlEncode(Session("user").ToString())
        Dim cookie As HttpCookie = Request.Cookies("user")
        lblCookie.Text = If(cookie Is Nothing,
            "(not found)", Server.HtmlEncode(Server.UrlDecode(cookie.Value)))
    End Sub

    Protected Sub btnLogout_Click(sender As Object, e As EventArgs)
        Session.Clear()
        Session.Abandon()
        Dim expired As New HttpCookie("user", "")
        expired.Expires = DateTime.Now.AddDays(-1)
        expired.HttpOnly = True
        expired.Secure = Request.IsSecureConnection
        Response.Cookies.Add(expired)
        Response.Redirect("LoginSession.aspx")
    End Sub
End Class
```

Common project files (reproduced in the appendix): site.css, Web.config.

EXPECTED OUTPUT

```
Login -> Welcome Page
From Session: Prince
From Cookie: Prince
Logout -> LoginSession.aspx; Session cleared.
```

Illustrative expected output based on source; Web Forms/LocalDB runtime testing was not possible on this host.

Note: The username cookie is never used as proof of authentication. This remains an instructional login sample, not a production authentication scheme.

APPENDIX

Shared Project Files

The following files are required by multiple programs. Each is printed in full here once; the respective practicals list the files they use.

COMMON FILE / App_Code/Db.vb

Used by: Assignment 2: 01, Assignment 2: 04, Assignment 2: 05, Assignment 2: 06, Assignment 2: 07

```
Imports System.Configuration
Imports System.Data
Imports System.Data.SqlClient

Public NotInheritable Class Db
    Private Sub New()
    End Sub

    Private Shared ReadOnly ConnectionText As String =
        ConfigurationManager.ConnectionStrings("con").ConnectionString

    Public Shared Function ReadTable(query As String,
                                     ParamArray args() As SqlParameter) As DataTable
        Dim result As New DataTable()
        Using con As New SqlConnection(ConnectionText)
            Using cmd As New SqlCommand(query, con)
                If args.Length > 0 Then cmd.Parameters.AddRange(args)
                Using adapter As New SqlDataAdapter(cmd)
                    adapter.Fill(result)
                End Using
            End Using
        End Using
        Return result
    End Function

    Public Shared Function Execute(query As String,
                                   ParamArray args() As SqlParameter) As Integer
        Using con As New SqlConnection(ConnectionText)
            Using cmd As New SqlCommand(query, con)
                If args.Length > 0 Then cmd.Parameters.AddRange(args)
                con.Open()
                Return cmd.ExecuteNonQuery()
            End Using
        End Using
    End Function
End Class
```

COMMON FILE / site.css

Used by: Assignment 2: 01, Assignment 2: 02, Assignment 2: 03, Assignment 2: 04, Assignment 2: 05, Assignment 2: 06, Assignment 2: 07, Assignment 2: 08, Assignment 2: 09, Assignment 2: 10

```
body { margin:0; background:#f4f8fa; color:#17344a;
    font:15px/1.55 Arial,sans-serif; }
.wrapper { max-width:930px; margin:34px auto; padding:26px 30px;
    border-radius:11px; box-shadow:0 3px 24px #17344a17; background:#fff; }
h1,h2 { color:#113249; }
.field { margin:9px 0; display:flex; gap:15px; align-items:center; }
.field label { width:160px; font-weight:bold; }
input,select,textarea { padding:9px 10px; border:1px solid #b9cdd6;
    border-radius:6px; min-width:230px; box-sizing:border-box; }
.action { background:#0a6774; border:0; border-radius:5px;
    color:#fff; padding:10px 16px; cursor:pointer; margin:6px 8px 6px 0; }
.grid { border-collapse:collapse; width:100%; margin-top:20px; }
.grid th { background:#e6f1f3; color:#133047; }
.grid td,.grid th { border-bottom:1px solid #d6e1e7;
    text-align:left; padding:9px; }
.message { margin:11px 0; display:block; color:#0a6774; }
.error { color:#b42318; }
.portal-head { background:#123047; color:#fff; padding:16px 24px; }
.portal-head h1 { color:#fff; margin:0; }
nav { background:#e5f1f3; padding:12px 24px; }
```

```
nav a { padding-right:20px; text-decoration:none; color:#0a6774; }
footer { color:#526475; padding:15px 24px; background:#e5f1f3; }
@media(max-width:650px) { .wrapper{margin:8px;padding:15px}
.field{display:block}.field label{display:block;width:auto} }
```

COMMON FILE / style.css

Used by: Basic Asp.Net Practicals: 01, Basic Asp.Net Practicals: 02, Basic Asp.Net Practicals: 03, Basic Asp.Net Practicals: 04, Basic Asp.Net Practicals: 05, Basic Asp.Net Practicals: 06, Basic Asp.Net Practicals: 07, Basic Asp.Net Practicals: 08, Basic Asp.Net Practicals: 09, Basic Asp.Net Practicals: 10

```
body { margin:0; font:16px/1.55 Arial,sans-serif; color:#17344a; background:#f5f8fa; }
main { max-width:760px; margin:46px auto; padding:28px; border-radius:12px;
  background:white; box-shadow:0 4px 24px #12304718; }
h1 { margin-top:0; color:#123047; font-size:27px; }
label { display:block; margin:12px 0 5px; font-weight:600; }
input, select { padding:10px; border:1px solid #b7c9d0; border-radius:6px; }
button, input[type=submit] { background:#0a6774; color:white; border:0;
  border-radius:6px; padding:10px 17px; cursor:pointer; margin:10px 0; }
.message { background:#edf7f6; padding:12px; display:block; margin-top:14px; }
.error { color:#b42318; }
```

COMMON FILE / Web.config

Used by: Assignment 2: 01, Assignment 2: 02, Assignment 2: 03, Assignment 2: 04, Assignment 2: 05, Assignment 2: 06, Assignment 2: 07, Assignment 2: 08, Assignment 2: 09, Assignment 2: 10

```
<?xml version="1.0" encoding="utf-8"?>
<configuration>
  <appSettings>
    <add key="ValidationSettings:UnobtrusiveValidationMode" value="None" />
  </appSettings>
  <connectionStrings>
    <add name="con" providerName="System.Data.SqlClient"
      connectionString="Data Source=(LocalDB)\MSSQLLocalDB;Initial Catalog=PracticalDB;Integrated
Security=True" />
  </connectionStrings>
  <system.web>
    <compilation debug="true" targetFramework="4.7.2" />
    <httpRuntime targetFramework="4.7.2" />
  </system.web>
</configuration>
```